

Weekly Report

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1 Progress

- I tested where the uncertain parts are located in the image through experiments $\rightarrow$ I confirmed that they gather in the high-frequency (detailed) areas.
- I compared different ways to measure uncertainty $\rightarrow$ "Temporal Uncertainty" is better because it costs less to calculate.
- I tested if adding "Importance" using a Vision Transformer is effective $\rightarrow$ Right now, I do not see enough improvement in accuracy.

2 System Model: Uncertainty-Aware Retransmission Diffusion Denoising Model

In this section, I explain the details of my proposed method, the "Uncertainty-Aware Retransmission Diffusion Denoising Model (U-RDDM)." This method is based on "DiffCom," a communication method that uses generative models. I measure the "uncertainty" that happens when the diffusion model restores the image. Based on this uncertainty map, I send the data again (retransmit). This makes the system stronger against noise in the communication channel.

2.1 Basic Model: DiffCom

In this study, I use DiffCom as the base framework. DiffCom uses a trained diffusion model as a receiver (decoder). It restores the image by sampling from the posterior distribution, using the received signal as a guide.

On the sender side, I change the input image $\mathbf{x} \in \mathbb{R}^{C \times H \times W}$ into channel input symbols $\mathbf{z} = \mathcal{E}(\mathbf{x})$ using a deterministic encoder $\mathcal{E}$ like DeepJSCC. The signal $\mathbf{y}$ received after passing through the wireless channel $\mathcal{W}$ is written as:

$$\mathbf{y} = \mathcal{W}(\mathbf{z}) + \mathbf{n} \quad (1)$$

Here, $\mathcal{W}(\cdot)$ represents channel distortions like fading, and $\mathbf{n}$ is additive Gaussian noise.

The DiffCom receiver uses the received signal $\mathbf{y}$ as a guide. It estimates the sent image $\mathbf{x}$ through the reverse diffusion process. Specifically, at each time step t of the diffusion model, I approximate the posterior score function $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y})$ and sample it as follows:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{y}) \approx s_\theta(\mathbf{x}_t, t) - \zeta_t \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{W}(\mathcal{E}(\hat{\mathbf{x}}_{0|t}))\|_2^2 \quad (2)$$

Here, $s_\theta(\mathbf{x}_t, t)$ is the score estimated by the pre-trained unconditional diffusion model, and ζ_t is a number that controls how strong the guidance is. Also, $\hat{\mathbf{x}}_{0|t}$ is the estimated value of x_0 predicted from the current $\mathbf{x}_t$ using Tweedie's formula. The second term is a gradient term to ensure "Measurement Consistency." It checks the difference between the current estimated image (passed through the channel again) and the actual received signal $\mathbf{y}$. This allows image generation that fits the received signal.

2.1.1 Temporal Uncertainty

Temporal Uncertainty $\mathbf{U}_{\text{temp}}$ is a score that checks if the predictions are consistent (steady) throughout all reverse diffusion steps $t \in \{T, T-1, \dots, 1\}$. Let $\mathcal{S} = \{\hat{\mathbf{x}}_{0|t}\}_{t=1}^T$ be the series of estimated images at each step. Temporal Uncertainty is defined as the variance along the time axis.

$$\mathbf{U}_{\text{temp}}^{(h,w)} = \frac{1}{C} \sum_{c=1}^C \text{Var}_{t \in \{1, \dots, T\}} [\hat{\mathbf{x}}_{0|t}^{(c,h,w)}] \quad (3)$$

Here, $\text{Var}_t[\cdot]$ represents the sample variance over the time steps.

2.1.2 Perturbation Uncertainty [2]

Perturbation Uncertainty $\mathbf{U}_{\text{pert}}$ is a score that evaluates local stability at a specific time step t . For the current estimate $\hat{\mathbf{x}}_{0|t}$, I use the Monte Carlo method to resample (perturb) it M times. The perturbed state $\mathbf{x}_t^{(k)}$ in the k -th trial ($k = 1, \dots, M$) is generated based on the definition of the diffusion process:

$$\mathbf{x}_t^{(k)} = \sqrt{\bar{\alpha}_t} \hat{\mathbf{x}}_{0|t} + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}^{(k)}, \quad \boldsymbol{\epsilon}^{(k)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (4)$$

Here, $\bar{\alpha}_t$ is the noise scale factor. Using this $\mathbf{x}_t^{(k)}$, I let the model estimate x_0 again, calling it $\hat{\mathbf{x}}_0^{(k)} = f_{\theta}(\mathbf{x}_t^{(k)}, t)$. Perturbation Uncertainty is defined as the variance of these M predicted values.

$$\mathbf{U}_{\text{pert}}^{(h,w)} = \frac{1}{C} \sum_{c=1}^C \text{Var}_{k \in \{1, \dots, M\}} [\hat{\mathbf{x}}_0^{(k)(c,h,w)}] \quad (5)$$

According to De Vita et al. [2], the variance of these estimates is related to the second derivative (Hessian matrix) of the log-likelihood of the noise distribution $\log q(\mathbf{x}_t)$. Specifically, a large variance means the curvature of the log-likelihood is high. This suggests it is an unstable area where the restoration result reacts sensitively to small changes in the input.

The original paper only considered uncertainty at a specific time step and did not consider uncertainty across the whole reverse process. Therefore, I calculate the final uncertainty map $\mathbf{U}$ by averaging the $\mathbf{U}_t$ created at specific time steps.

This week, I compared which contributes more to performance improvement: local instability or global instability over the entire restoration path.

2.2 Semantic Retransmission

Based on the calculated uncertainty map $\mathbf{U}$, the receiver decides which signal areas to send again. Furthermore, to consider "Structural Relevance" (importance), I introduce a Saliency Map $\mathbf{A}$ using a ViT (Vision Transformer).

The Saliency Map $\mathbf{A}$ is generated as follows. Let N_h be the number of heads in the final layer of ViT, and $w_{i,j}$ be the Attention weight from the [CLS] token to the j -th image patch token in head i . The average importance vector $\mathbf{v} \in \mathbb{R}^P$ (P is the total number of patches) across all heads is calculated by:

$$\mathbf{v}_j = \frac{1}{N_h} \sum_{i=1}^{N_h} w_{i,j} \quad (6)$$

I reshape this vector $\mathbf{v}$ into a 2D grid matching the patch layout and resize it to the input image size (H, W) using Bilinear Interpolation to get the unnormalized map $\mathbf{A}_{\text{raw}}$. The final Saliency Map $\mathbf{A}$ is obtained by Min-Max normalizing this to the range $[0, 1]$.

$$\mathbf{A} = \frac{\mathbf{A}_{\text{raw}} - \min(\mathbf{A}_{\text{raw}})}{\max(\mathbf{A}_{\text{raw}}) - \min(\mathbf{A}_{\text{raw}}) + \epsilon} \quad (7)$$

Here, ϵ is a very small value to keep the calculation stable. Using these, I define the Retransmission Priority Map $\mathbf{P}$ as follows:

$$\mathbf{P} = \mathbf{U} \odot \mathbf{A} \quad (8)$$

Here, $\odot$ represents the element-wise product. A region with high priority $\mathbf{P}$ means it is both hard to restore and perceptually important.

The retransmission mask $\mathbf{M} \in \{0, 1\}^{C' \times H' \times W'}$ is created by downsampling the priority $\mathbf{P}$ to the size of the latent signal output by the encoder. I assign 1 to the top $K\%$ regions or regions above a threshold. For the received signal $\mathbf{y}_{\text{dirty}}$, I use the retransmitted clean signal $\mathbf{y}_{\text{clean}}$ to perform Signal Replacement:

$$\mathbf{y}_{\text{new}} = (\mathbf{1} - \mathbf{M}) \odot \mathbf{y}_{\text{dirty}} + \mathbf{M} \odot \mathbf{y}_{\text{clean}} \quad (9)$$

Using this updated observation signal $\mathbf{y}_{\text{new}}$ as a new condition, I perform the second stage of diffusion restoration. This maximizes perceptual quality while keeping data traffic low.

3 Experiments

3.1 Experimental Settings

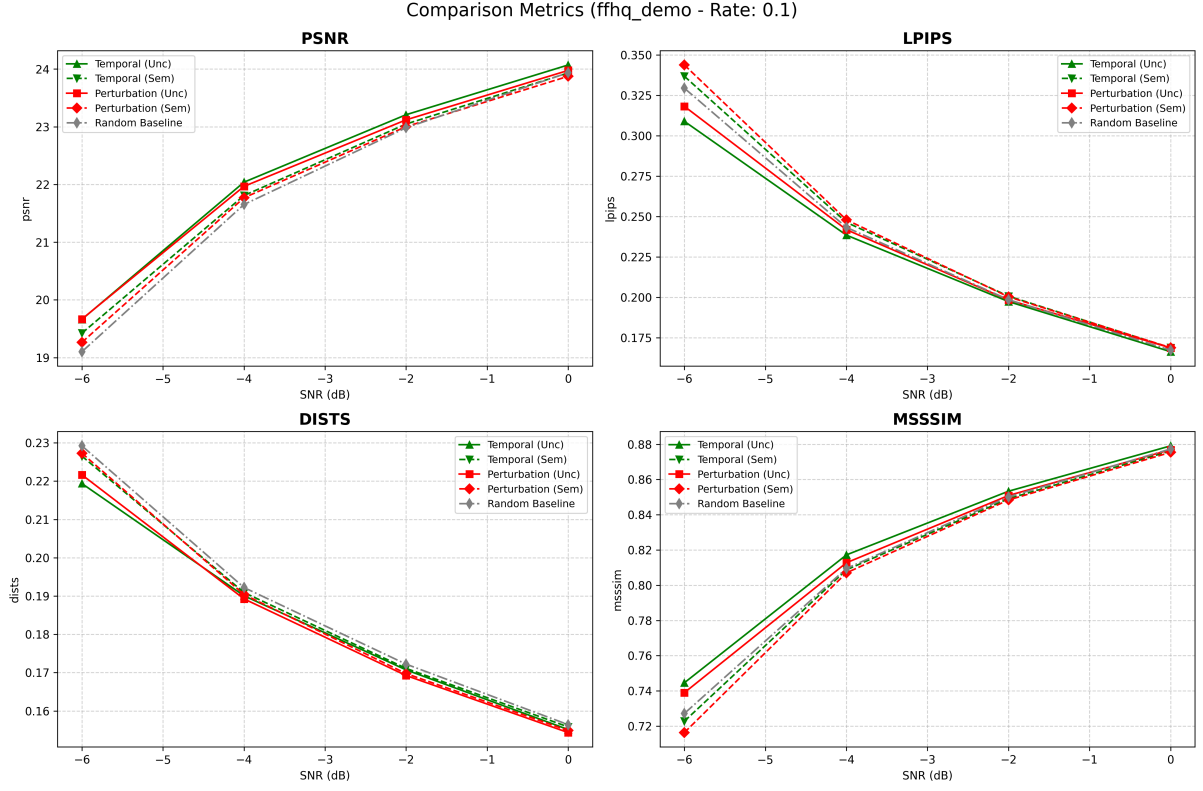
Table 1: Experimental Settings

Item	Setting
Diffusion Model	DDPM
Encoder	DeepJSCC
ViT Model	DINOv2 (Small)
Dataset	FFHQ, ImageNet 256×256
ζ	0.3
Channel	AWGN

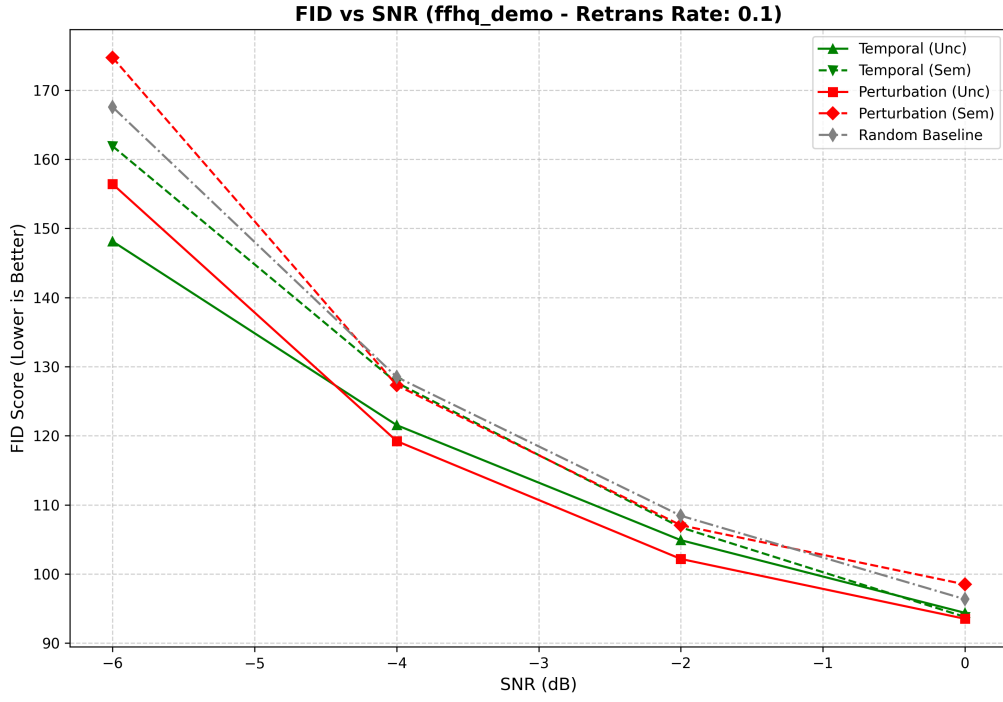
The methods I compared are as follows:

- **Temporal (Unc)** : Create mask using only Temporal Uncertainty.
- **Temporal (Sem)** : Semantic Retransmission using Temporal Uncertainty and Structural Relevance.
- **Perturbation (Unc)** : Create mask using only Perturbation Uncertainty.
- **Perturbation (Sem)** : Semantic Retransmission using Perturbation Uncertainty and Structural Relevance.
- **Random Baseline** : Create retransmission mask randomly.

3.2 Experimental Results

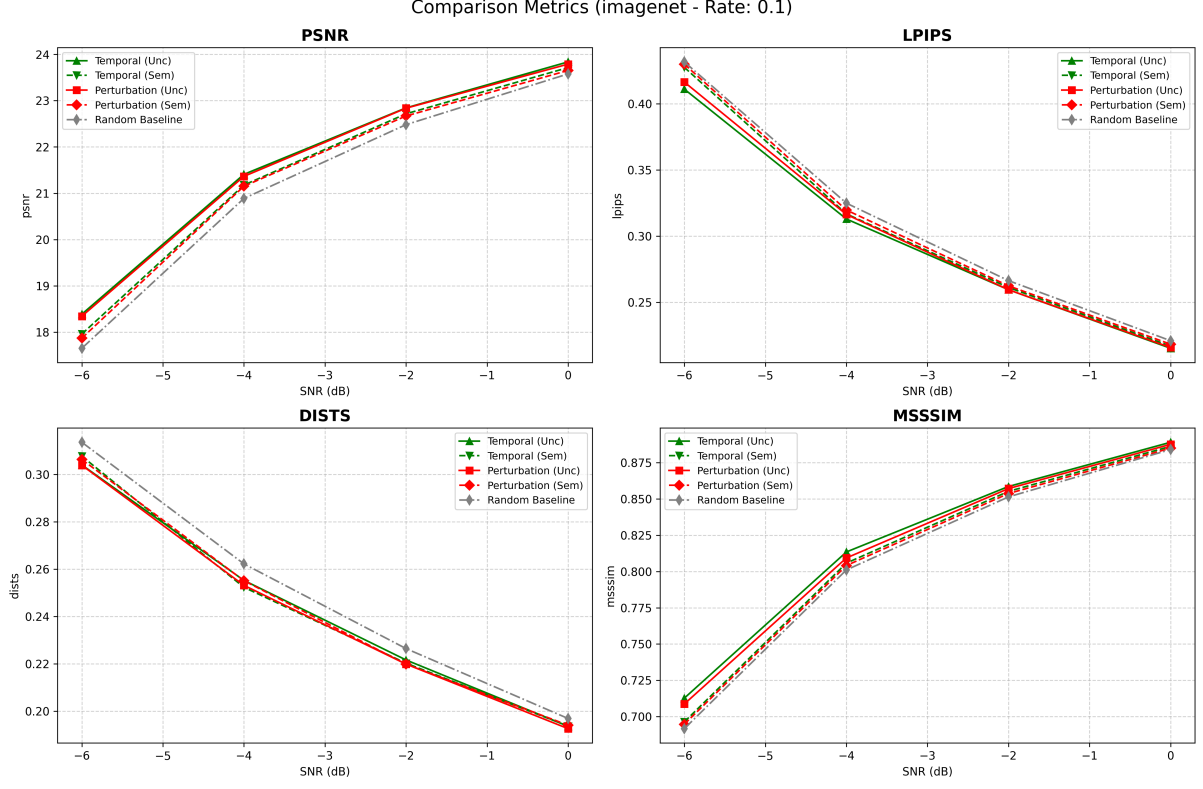


(a) Metrics vs SNR

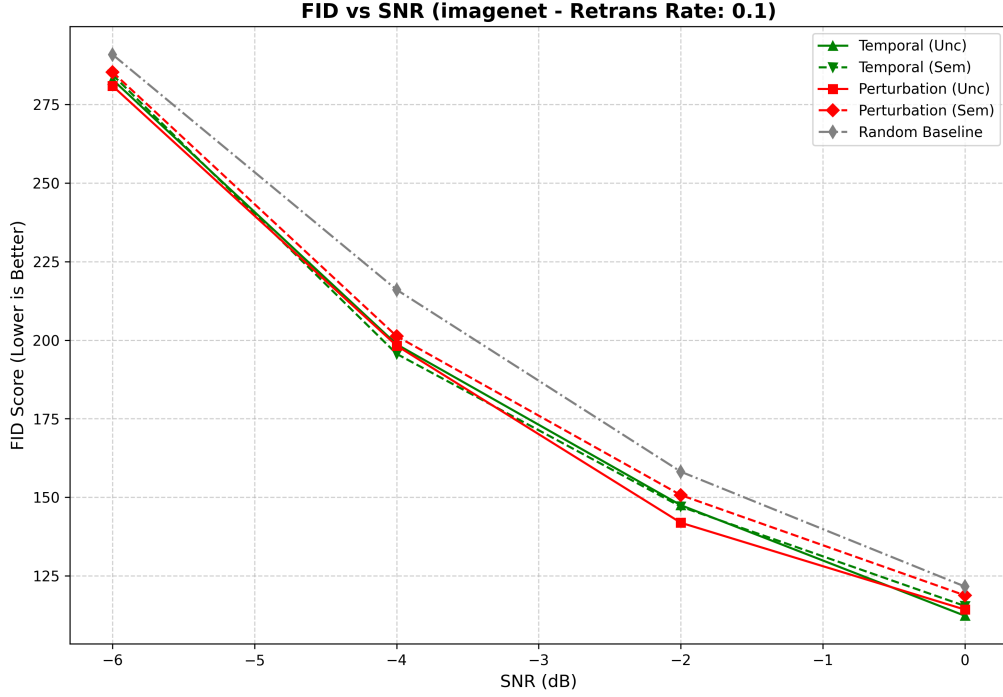


(b) FID vs SNR

Figure 1: SNR characteristics of image quality metrics on the FFHQ dataset



(a) Metrics vs SNR



(b) FID vs SNR

Figure 2: SNR characteristics of image quality metrics on the ImageNet dataset

From the experimental results, in the low SNR (< 0 dB) range, the proposed method using uncertainty showed better restoration accuracy (like PSNR) compared to the Random Baseline. On the other hand, in the high SNR (> 0 dB) range, the Random Baseline tended to be better

in PSNR, LPIPS, and MSSIM metrics.

However, regarding FID, the advantage of the method using uncertainty was clear across all SNR ranges. This suggests that focusing on the uncertainty of the diffusion model contributes greatly to keeping the generated images natural. When the diffusion model is unstable (unsure) during the generation process, it tends to create artifacts or "hallucinations" (seeing things that aren't there) while trying to force consistency with the surroundings. I confirmed that retransmission using my proposed method efficiently stops this unnatural generation.

Next, I perform a numerical evaluation. Below are the results when SNR = -4, -6 dB, where the difference from the Random Baseline was most visible.

Table 2: Results at SNR=-4dB

Method	PSNR	MSSSIM	LPIPS	DISTS	FID	HyperIQA
First Restoration	20.96	0.785	0.268	0.202	138.9	0.6815
Random Baseline	21.65	0.810	0.243	0.192	128.5	0.6855
Temporal (Unc)	22.04	0.817	0.239	<u>0.190</u>	<u>121.5</u>	0.6934
Temporal (Sem)	21.83	0.809	0.246	0.191	127.7	0.6905
Perturbation (Unc)	<u>21.97</u>	<u>0.813</u>	<u>0.242</u>	0.189	119.2	<u>0.6925</u>
Perturbation (Sem)	21.76	0.807	0.248	<u>0.190</u>	127.3	0.6917

Table 3: Results at SNR=-6dB

Method	PSNR	MSSSIM	LPIPS	DISTS	FID	HyperIQA
First Restoration	17.90	0.667	0.396	0.260	209.5	0.6544
Random Baseline	19.10	0.727	0.329	0.229	167.6	0.6608
Temporal (Unc)	<u>19.66</u>	0.745	0.309	0.219	148.1	0.6678
Temporal (Sem)	19.42	0.723	0.337	0.227	161.9	0.6593
Perturbation (Unc)	19.67	<u>0.739</u>	<u>0.318</u>	<u>0.222</u>	<u>156.4</u>	<u>0.6667</u>
Perturbation (Sem)	19.27	0.716	0.344	0.227	174.7	0.6613

Also, between the two types of uncertainty I compared (Temporal and Perturbation), there was no big difference in performance, but Temporal was slightly better. However, Perturbation has a problem of high calculation cost because it requires the diffusion model to run inference multiple times at the same step. Therefore, I plan to adopt Temporal (Unc) as the proposed method for now.

In addition, I confirmed that Semantic Retransmission (Sem) was worse than using only the uncertainty map (Unc) in all metrics. A reason for this is that the importance estimation by ViT is easily affected by noise. Actually, looking at Fig.5, areas that are not related to the subject, like the top-left or bottom-right corners of the image, are evaluated as having high importance. Also, a main reason for the lower accuracy of Sem might be that the retransmission resources are too concentrated in one local area. I guess that to improve the overall restoration quality of the diffusion model, a strategy that spreads resources over a wider area is more effective.



Figure 3: Ground Truth (Sent Image)



Figure 4: Phase 1: 1st Restoration Image

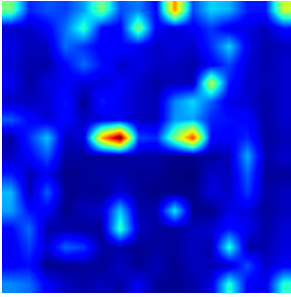


Figure 5: ViT Saliency Map (A)



Figure 6: Mask: Temporal (Unc)



Figure 7: Mask: perturbation (Unc)



Figure 8: Mask: Temporal (Sem)

Figure 9: Visualization of initial restoration image, and retransmission masks based on uncertainty and importance.



Figure 10: Random Baseline



Figure 11: Temporal (Unc)



Figure 12: Perturbation (Unc)



Figure 13: Temporal (Sem)

Figure 14: Comparison of final restoration results after retransmission ($\text{SNR} = -4\text{dB}$). The proposed methods (middle/right) reduce unnaturalness in eyes and hair flow compared to Random (left).

4 Next Step

- Writing my graduation thesis.
- Re-examining the Importance estimation method (Thinking of a method other than ViT).

References

- [1] S. Wang, J. Dai, K. Tan, X. Qin, K. Niu and P. Zhang, "DiffCom: Channel Received Signal Is a Natural Condition to Guide Diffusion Posterior Sampling," in *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 7, pp. 2651-2666, July 2025, doi: 10.1109/JSAC.2025.3559158.
- [2] M. De Vita and V. Belagiannis, "Diffusion Model Guided Sampling with Pixel-Wise Aleatoric Uncertainty Estimation," 2025 *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Tucson, AZ, USA, 2025, pp. 3844-3854, doi: 10.1109/WACV61041.2025.00378.